# torch.cuda.memory.host_memory_stats#

https://pytorch.org/docs/stable/generated/torch.cuda.memory.host_memory_stat

**Description:**

Return a dictionary of CUDA memory allocator statistics for a given device. The return value of this function is a dictionary of statistics, each of which is a non-negative integer. "allocated.{current,peak,allocated,freed}": number of allocation requests received by the memory allocator. "allocated_bytes.{current,peak,allocated,freed}": amount of allocated memory.

## Function Signature

```
torch.cuda.memory.host_memory_stats()[source]#
Return type
```

**Returns:**

dict[str, Any]

# Detailed Documentation

## Documentation

Return a dictionary of CUDA memory allocator statistics for a given device.

The return value of this function is a dictionary of statistics, each of which is a non-negative integer.

"allocated.{current,peak,allocated,freed}": number of allocation requests received by the memory allocator.

"allocated_bytes.{current,peak,allocated,freed}": amount of allocated memory.

"segment.{current,peak,allocated,freed}": number of reserved segments from cudaMalloc().

"reserved_bytes.{current,peak,allocated,freed}": amount of reserved memory.

For these core statistics, values are broken down as follows.

current: current value of this metric.

peak: maximum value of this metric.

allocated: historical total increase in this metric.

freed: historical total decrease in this metric.

In addition to the core statistics, we also provide some simple event counters:

"num_host_alloc": number of CUDA allocation calls. This includes both cudaHostAlloc and cudaHostRegister.

"num_host_free": number of CUDA free calls. This includes both cudaHostFree and cudaHostUnregister.

Finally, we also provide some simple timing counters:

"host_alloc_time.{total,max,min,count,avg}": timing of allocation requests going through CUDA calls.

"host_free_time.{total,max,min,count,avg}": timing of free requests going through CUDA calls.

For these timing statistics, values are broken down as follows.

total: total time spent.

max: maximum value per call.

min: minimum value per call.

count: number of times it was called.

avg: average time per call.